

Md. Easir Rahat

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PORTFOLIO

CAREER OBJECTIVE

Mechanical Engineering graduate from Rajshahi University of Engineering & Technology (RUET), working at the intersection of deep learning and physical systems, from thermal comfort prediction to parameter-efficient continual learning for vision transformers, grounded by hands-on automotive automation experience with Team Annexe RUET. Seeking to pursue PhD research in machine learning, deep learning, and image processing applied to automation and thermal systems, building on a strong academic foundation in design, production planning, quality control, and HVAC engineering.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)

B.Sc. Engineering in Mechanical Engineering

Rajshahi, Bangladesh

Mar 2022 – Aug 2026

CGPA: 3.33/4.00

- **Undergraduate Thesis:** Development of an ANN-Based Indoor Thermal Comfort Prediction and Monitoring System. Advisor: [Dr. Mhia Md. Zaglul Shahadat](#), Professor, Dept. of ME, RUET.
- **Notable Courses:** Thermodynamics, Fluid Mechanics, Heat Transfer, Machine Design, Manufacturing Processes, Production Planning & Control, HVAC & Refrigeration, Control Systems, Industrial Engineering & Management.

New Govt. Degree College

Higher Secondary Certificate (HSC)

Rajshahi, Bangladesh

2018 – 2020

GPA: 5.00/5.00

Joypurhat R.B. Govt. High School

Secondary School Certificate (SSC)

Joypurhat, Bangladesh

2016 – 2018

GPA: 5.00/5.00

RESEARCH INTERESTS

- Machine Learning
- Image Processing
- Computational Fluid Dynamics (CFD)
- Deep Learning
- Automation
- Thermal Systems

PUBLICATIONS

Journal / Conference Articles

Submitted

- [S-1] **Md. Easir Rahat**, Md. Ferdaous Al-Farabe. “Development of an ANN-Based Indoor Thermal Comfort Prediction and Monitoring System.” Submitted to the 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox’s Bazar, IEEE Bangladesh Section. An ensemble artificial neural network predicting indoor thermal comfort in real time, benchmarked against Random Forest, XGBoost, and Linear Regression, validated against the CBE Thermal Comfort Tool, and deployed on embedded (ESP32) hardware with a live monitoring dashboard. [\[Link\]](#)
- [S-2] **Md. Easir Rahat**, Md. Tajbir Hasan Shuvo. “Revisiting Design Choices in LoRA-Based Continual Learning for Vision Transformers in Semantic Segmentation: An Empirical Study.” Submitted to the 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox’s Bazar. An empirical study reimplementing a LoRA-based continual learning pipeline (CLoRA) on a frozen ViT-B/16 backbone, systematically varying backbone pretraining, decoder depth, distillation weight, and LoRA rank across four PASCAL VOC splits to isolate which design choices drive performance and forgetting resistance. [\[Link\]](#)

RESEARCH & WORK EXPERIENCE

Industrial Trainee, Engineering Department

Square Pharmaceuticals PLC, Pabna Unit, Bangladesh

10 – 23 Mar 2025

ME 4120 Industrial Training; Supervisor: Md. Khairul Alam, DGM (Production Unit)

- Completed a two-week attachment rotating across the plant’s utility and support systems rather than a single production line: the HVAC system serving cleanrooms and warehouses (air handling units, chillers, dehumidifiers, and the Building Management System), the compressed air network (17 compressors across three formulation units), and the 11/0.4kV substation stepping down power from the plant’s 9MW captive gas-turbine plant.
- Studied the purified water and Water-for-Injection (WFI) systems and their chemical/microbial testing limits, along with the nitrogen and oxygen generation plants, which use molecular-sieve adsorption (CMS/GMS) rather than cryogenic separation.
- Observed Quality Control testing of raw materials, in-process samples, and finished products against BP/USP/EP standards, and the instrument calibration schedule; final report raised observations on manual packaging bottlenecks and warehouse capacity constraints.

PROJECTS

Academic Projects

Three-Axis Robotic Arm

- Designed and built a robotic arm with three-axis rotation using servo motors and an Arduino Uno, with each joint independently controlled through a dedicated push-button pair; full circuit design was built and tested on a breadboard before final assembly.

Object-Pulling Crane Prototype

- Developed a working crane prototype, applying core principles of mechanical linkages and load handling.

Mechanical Parts Design (AutoCAD)

- Modeled various mechanical parts in AutoCAD, gaining hands-on experience in 2D/3D drafting and engineering drawing.

Simulation Studies

Brake System Simulation (ANSYS Static Structural)

- Static structural analysis of a wheel brake/lever assembly under a 300 lbf braking load, evaluating equivalent (von-Mises) stress and displacement with fixed, frictionless, and pivot displacement boundary conditions.

Pipe Flow Simulation (SolidWorks Flow Simulation)

- Internal flow simulation of a multi-bend piping loop, analyzing fluid temperature, vorticity-colored flow trajectories, and boundary layer development through the bends and junctions.

Gear Contact Stress Analysis (ANSYS Static Structural)

- Static structural analysis of a meshing spur gear pair under 274 lbf-in applied torque, with a refined tooth-contact mesh to resolve local stress concentration, deformation, and contact behavior.

Team Project (Formula Kart Design Challenge, FKDC)

Go-Kart Quality Control, Team Annexe RUET

- Led quality control for the design and manufacturing of a Go-Kart competing at FKDC, Coimbatore, India: checked incoming materials and dimensions against design tolerances, tracked fit issues during welding and assembly, and sourced components on a tight budget without compromising safety-critical parts (steering, brakes, roll structure). ANSYS was used for structural checks on the chassis before fabrication.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Class Representative, Dept. of Mechanical Engineering, RUET

Oct 2023 – Aug 2026

- Served as the primary liaison between faculty and 180+ students across three sections for over two years; organized departmental farewell and reception ceremonies, class parties, and the industrial tour.

Vice President (Finance), ASHRAE RUET Student Branch

Jul 2025 – Aug 2026

- Coordinated two technical events, including a two-day HVAC Design seminar with 150+ participants held in collaboration with the ASHRAE Bangladesh Chapter, and managed financial planning and resource allocation for branch activities.

Vice President (Program), RUET Mechanical Club

2025 – 2026

- Organized the “Thesis Roadmap & Case Study Competition,” a department-wide event on research planning and applied case analysis.

General Secretary, Joypurhat Zilla Association of RUET

Jun 2025 – Jul 2026

- Organized Iftar parties, get-togethers, the annual picnic, and sports events for 50+ association members; managed admission booth arrangements and organized a Science Olympiad in Joypurhat.

Quality Control Team Lead, Team Annexe RUET

Aug 2025 – Aug 2026

- Led quality control for a Go-Kart competing in the Formula Kart Design Challenge, conducting materials selection, dimensional analysis, and market analysis to identify cost-effective, high-quality components.

SKILLS

- Technical:** Machine Design & Mechanical Drawing, Thermal & Fluid System Analysis, Manufacturing Process Planning, HVAC & Refrigeration System Design, CFD Fundamentals, Maintenance & Troubleshooting
- Quality Tools:** 5S, Six Sigma, Root Cause Analysis, 5M, Fishbone (Ishikawa) Diagram
- Software & ML:** AutoCAD, SolidWorks, ANSYS, MATLAB, PyTorch, LoRA / PEFT
- Productivity:** MS Excel (Formulas, Pivot Tables, Charts), MS PowerPoint (Technical Presentations)

REFERENCES

Dr. Md Rabiul Islam Sarker
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Dr. Mhia Md. Zaglul Shahadat
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